

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

اللَّهُمَّ إِنِّي أَسْأَلُكَ عِلْمًا نَافِعًا،

وَرِزْقًا طَيِّبًا، وَعَمَلًا مُتَقَبَّلًا،

Graphical Representation & Introduction to Probability

Week 3



Are People Happy?

North America



Career



Family



Security



Central America



Career



Family



Security



Europe



Career



Family



Security



Graphical Representation

- Scatter Plot
- Stem-and-Leaf Plot
- Histogram
- Box Plot

Scatter Plot

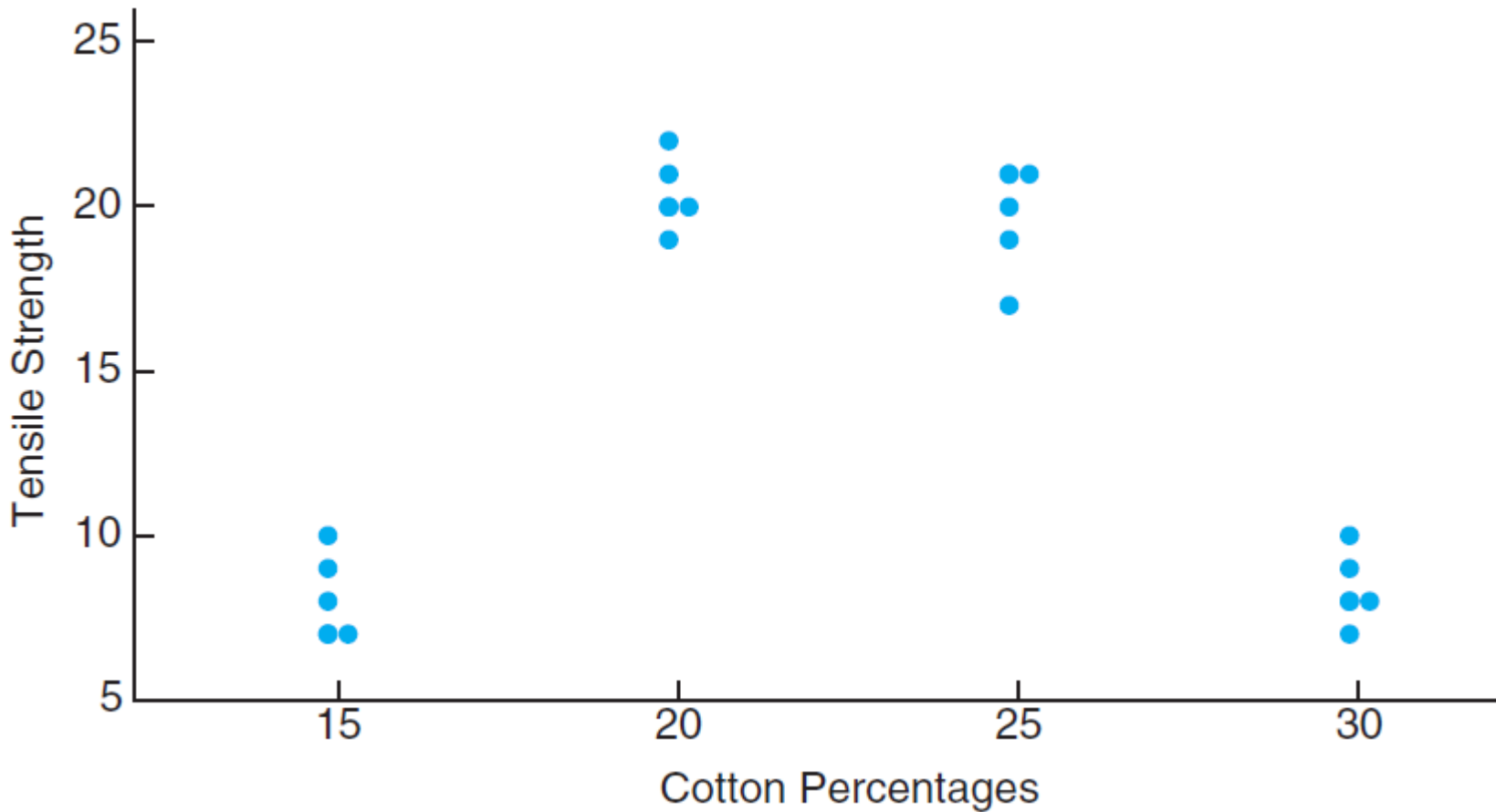


Table 1.3: Tensile Strength

Cotton Percentage	Tensile Strength
15	7, 7, 9, 8, 10
20	19, 20, 21, 20, 22
25	21, 21, 17, 19, 20
30	8, 7, 8, 9, 10

Stem and Leaf Plot

Table 1.4: Car Battery Life

2.2	4.1	3.5	4.5	3.2	3.7	3.0	2.6
3.4	1.6	3.1	3.3	3.8	3.1	4.7	3.7
2.5	4.3	3.4	3.6	2.9	3.3	3.9	3.1
3.3	3.1	3.7	4.4	3.2	4.1	1.9	3.4
4.7	3.8	3.2	2.6	3.9	3.0	4.2	3.5

Table 1.5: Stem-and-Leaf Plot of Battery Life

Stem	Leaf	Frequency
1	69	2
2	25669	5
3	0011112223334445567778899	25
4	11234577	8

Histogram

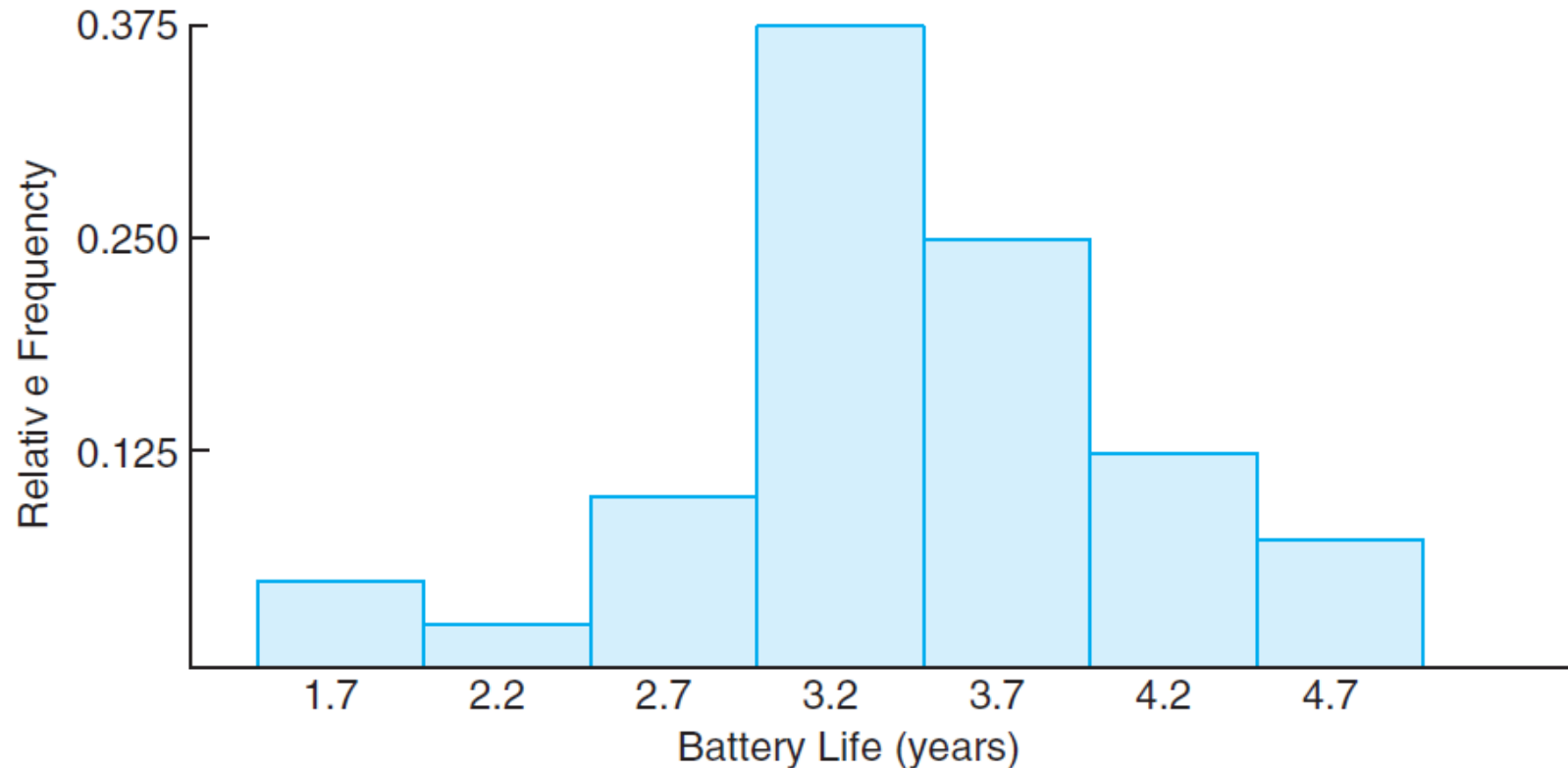
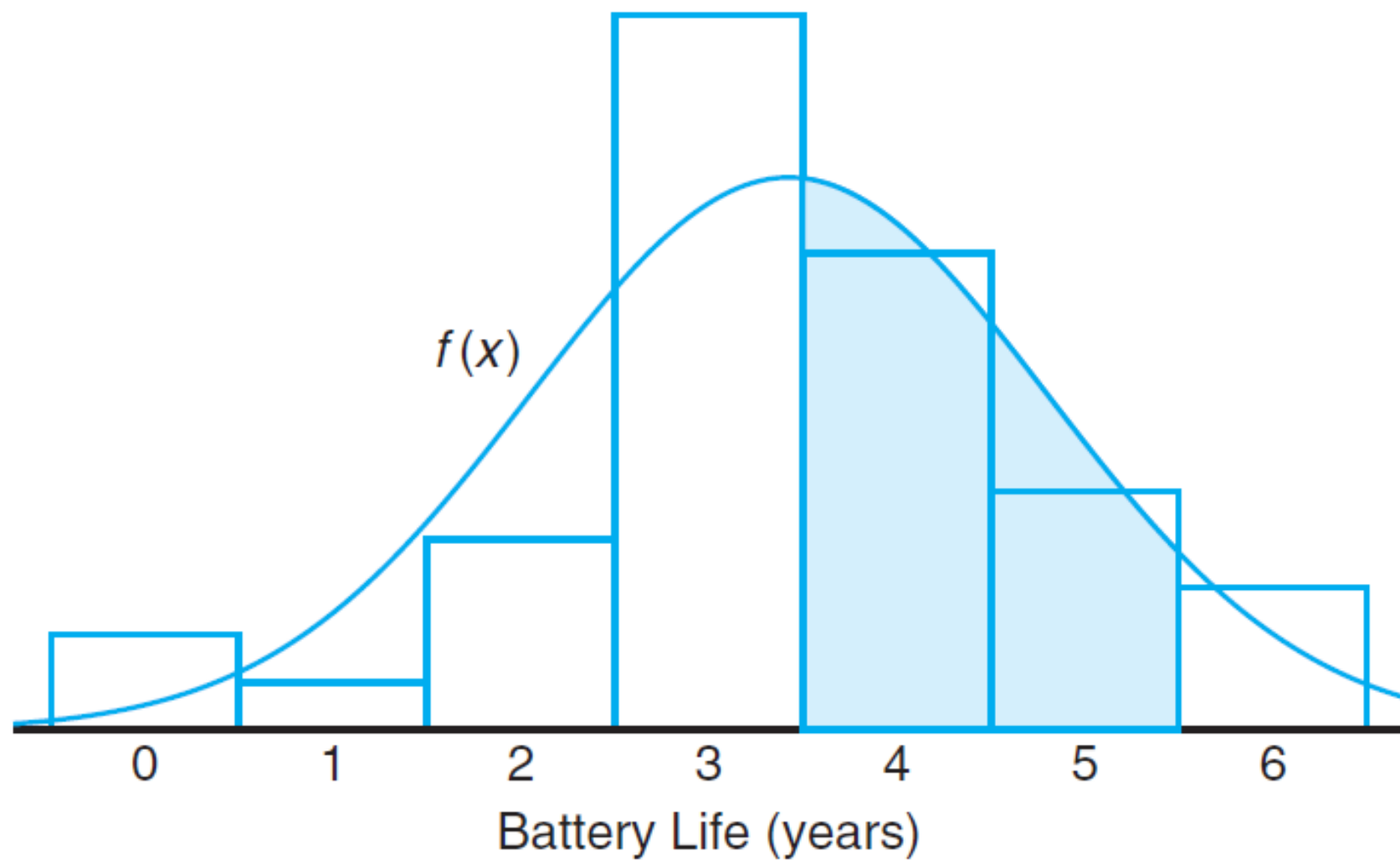


Table 1.7: Relative Frequency Distribution of Battery Life

Class Interval	Class Midpoint	Frequency, f	Relative Frequency
1.5–1.9	1.7	2	0.050
2.0–2.4	2.2	1	0.025
2.5–2.9	2.7	4	0.100
3.0–3.4	3.2	15	0.375
3.5–3.9	3.7	10	0.250
4.0–4.4	4.2	5	0.125
4.5–4.9	4.7	3	0.075



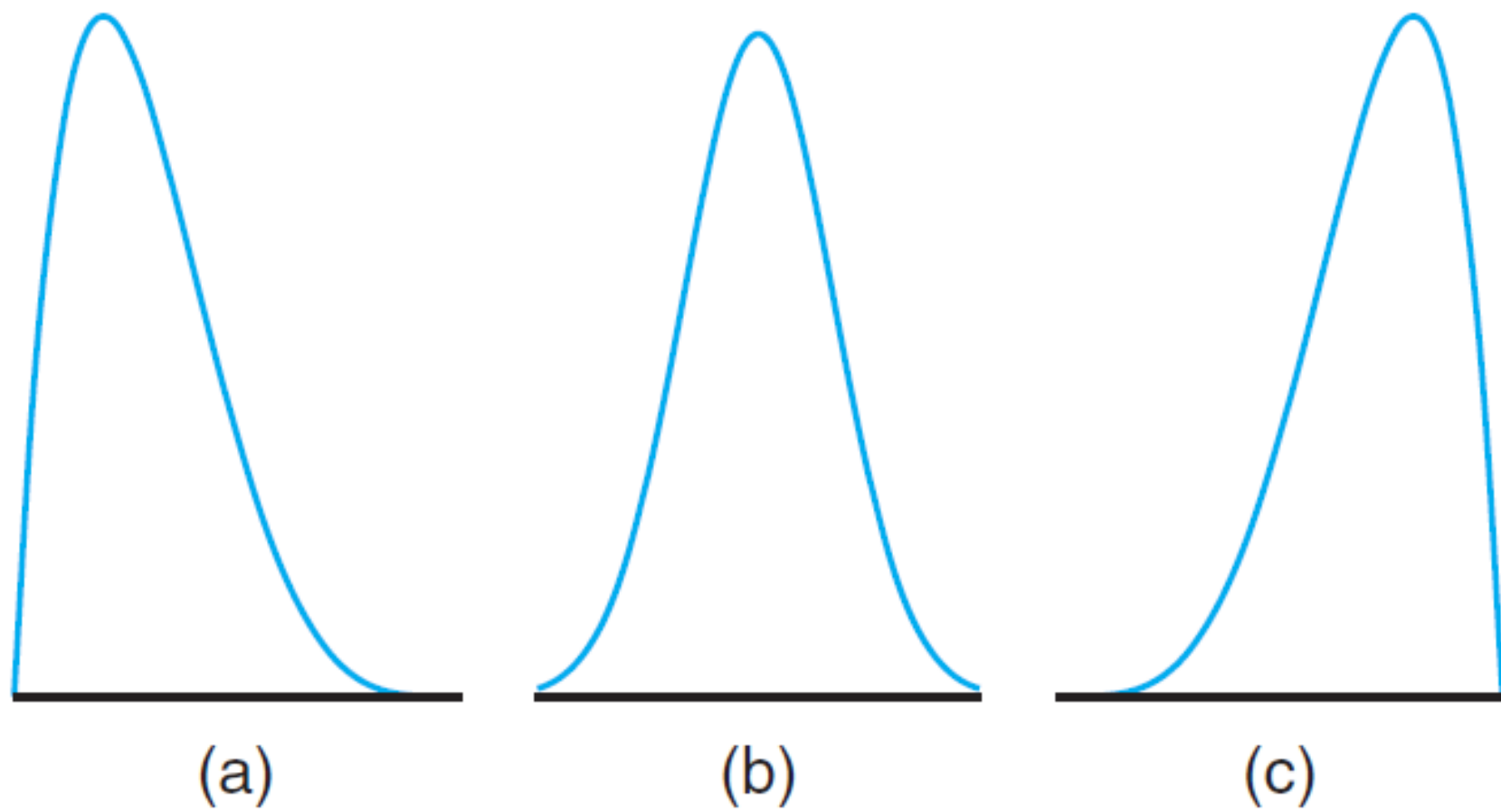
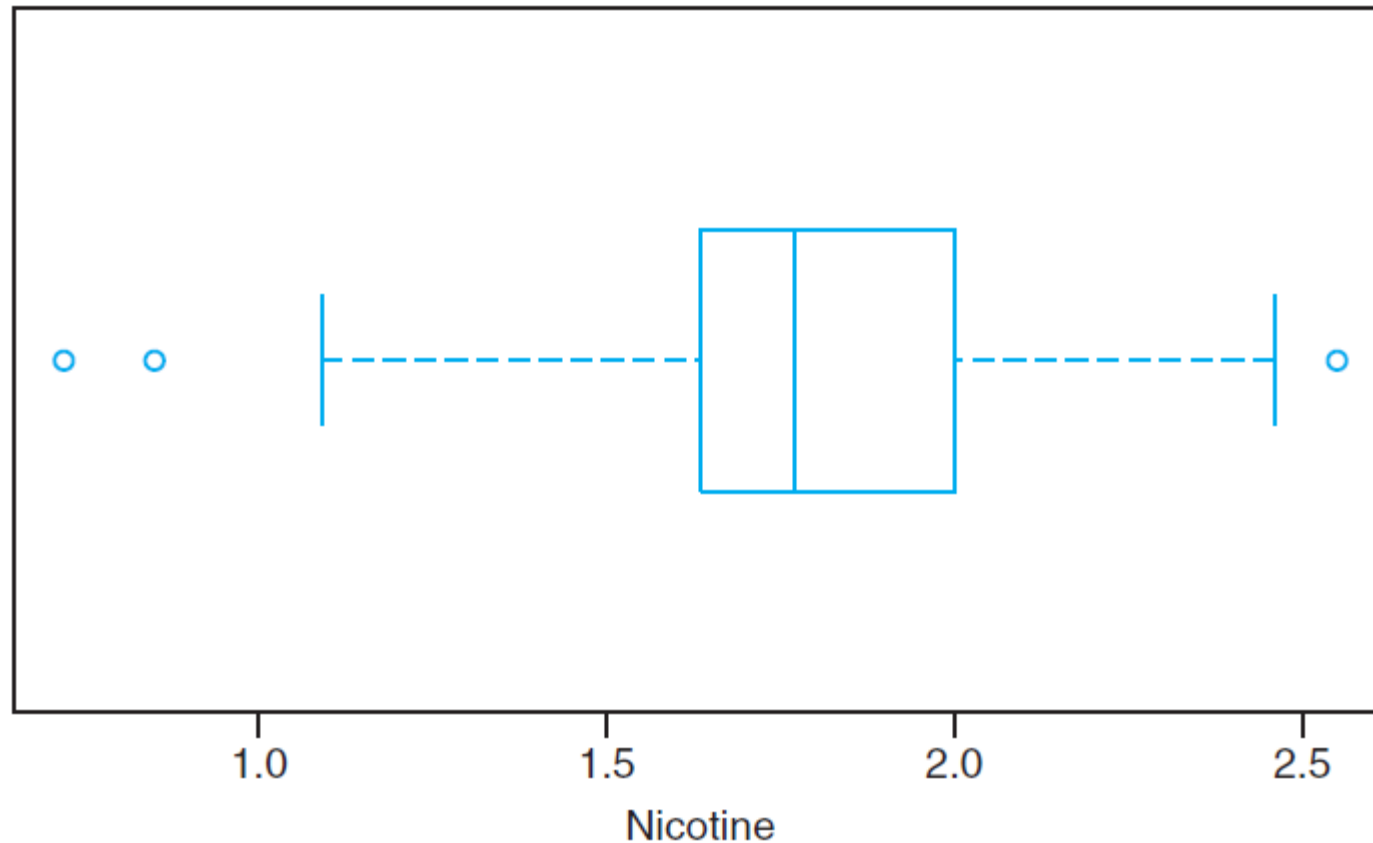


Figure 1.8: Skewness of data.

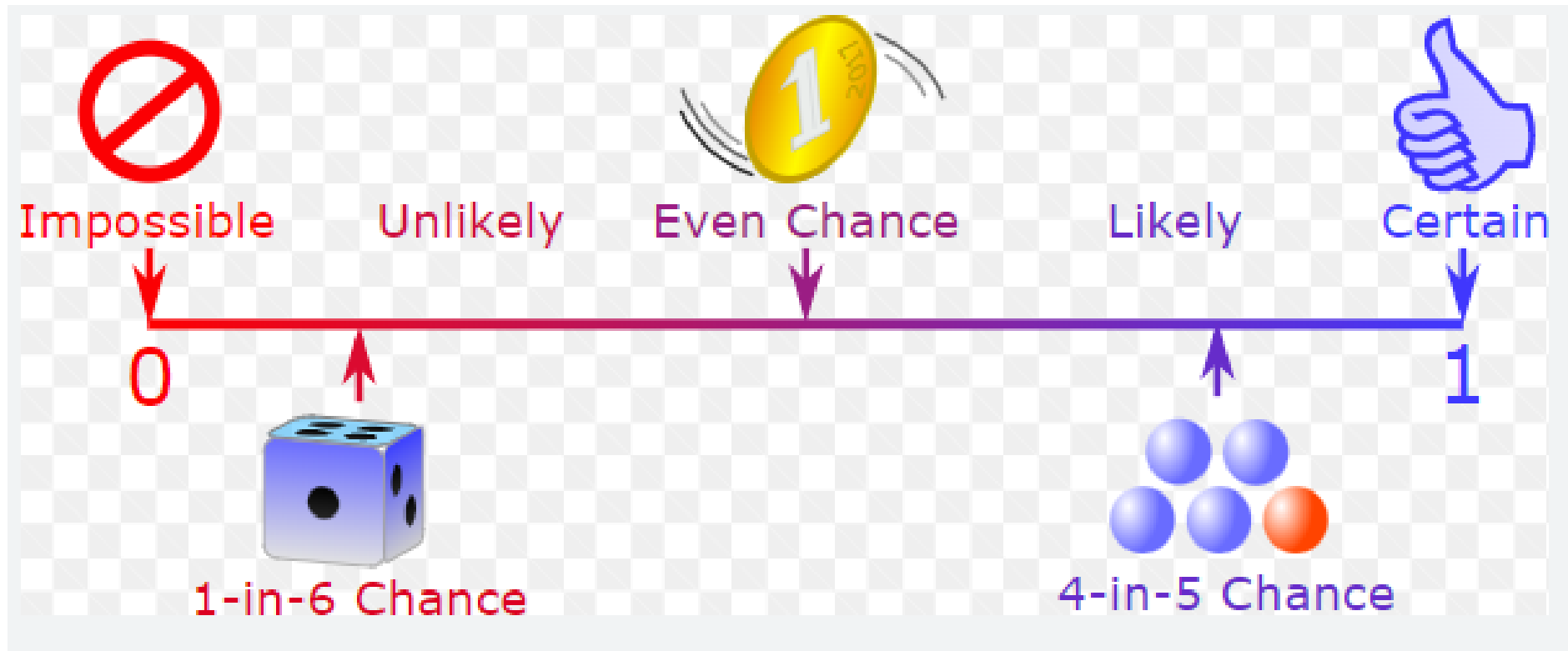
Box plot

Table 1.8: Nicotine Data for Example 1.5

1.09	1.92	2.31	1.79	2.28	1.74	1.47	1.97
0.85	1.24	1.58	2.03	1.70	2.17	2.55	2.11
1.86	1.90	1.68	1.51	1.64	0.72	1.69	1.85
1.82	1.79	2.46	1.88	2.08	1.67	1.37	1.93
1.40	1.64	2.09	1.75	1.63	2.37	1.75	1.69



Probability



Important Points

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(S) = 1$$

$$0 \leq P(A) \leq 1$$

$$P(\Phi) = 0$$

Role of Probability and Statistics in Engineering

Types of Uncertainties

1. Natural randomness (Aleatory)
2. Inaccuracies in prediction and estimation of reality
(Epistemic)

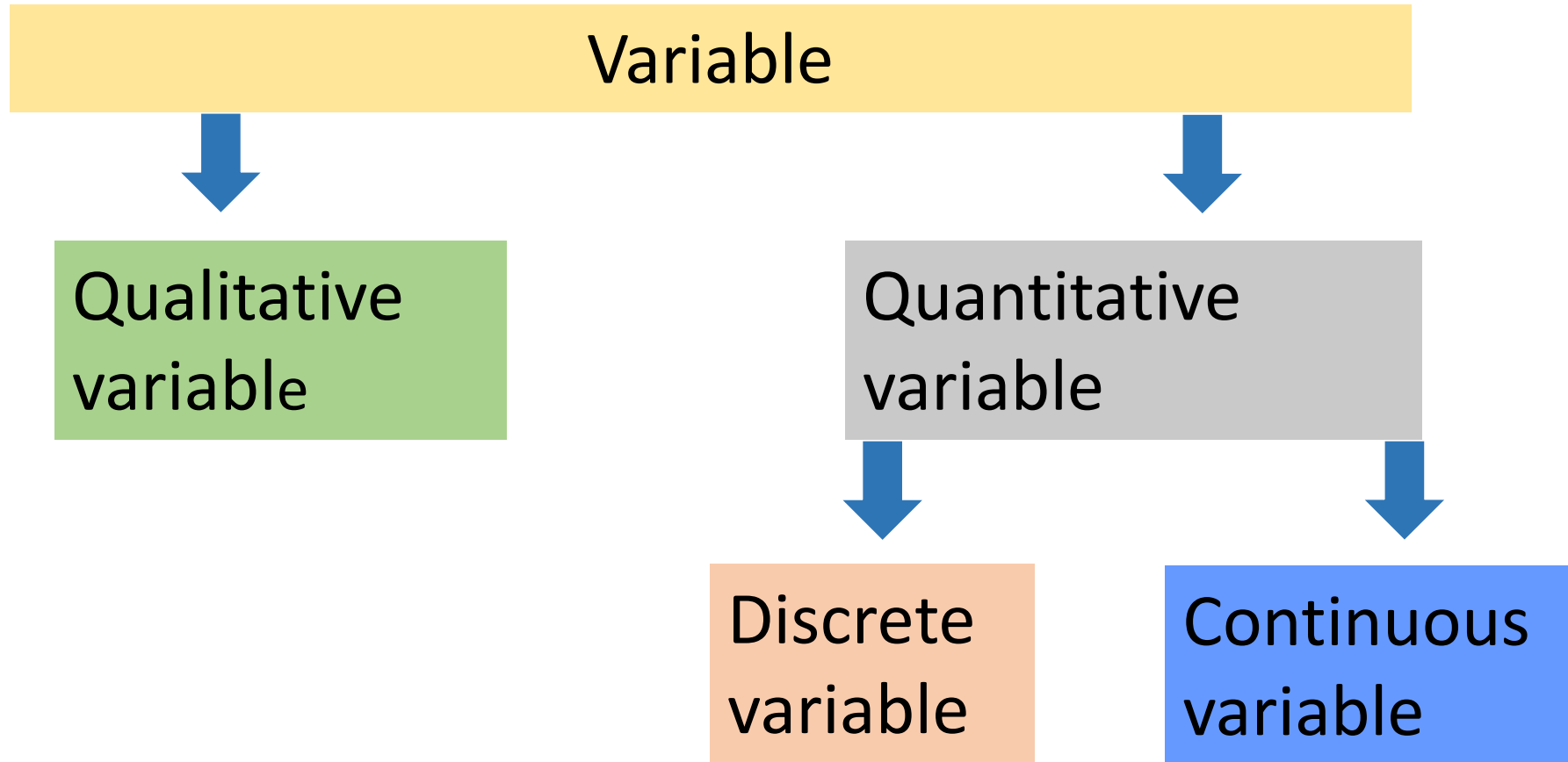
Experiment

Any process that generates a set of data.

Examples:

1. Tossing of a coin
2. Launching of a missile and observing of its velocity at specified times
3. During a survey, the opinions of voters concerning a new sales tax
4. In Pollution study, amount of Sulphur monoxide in air
5. Calculating the stresses and strains in structural members, such as beams, columns, and shafts

Types of Data



Examples for types of Data

1. Numerical

- i. Height/ Weight of students
- ii. Volume of gas
- iii. Number of accidents
- iv. Marks of students
- v. No of students in a department

2. Categorical

- i. Quality of product either defective/ non-defective
- ii. Color of eyes as blue, black, brown, grey,...
- iii. Color of hair of students as Black, rich brown, golden brown,...
- iv. Level of likeliness for chocolate/ candies/ ice-cream as most favorite, favorite, like, don't like, hate

Types of Data

1. The number of accidents that occur monthly at the U-turn in front of NUTECH, hoping to justify the installation of a traffic light
2. The volume of gas released in a chemical reaction when the concentration of an acid is varied
3. Items coming off an assembly line as defective or non-defective
4. Color of eyes in a graduate class at NUTECH
5. Marks attained by students in Mid-Term Exam
6. During a survey, the opinions of voters concerning a new sales tax
7. In Pollution study, amount of Sulphur monoxide in air
8. Calculating the stresses and strains in structural members, such as beams, columns, and shafts

Terms

Each outcome in a sample space is called an **element** or a **member** of the sample space, or simply a **sample point**.

Sample Space: The set of all possible outcomes of a statistical experiment is called the sample space and is represented by the symbol **S** .

Example:

Experiment: Toss a coin

Sample Point: H, T

Sample Space: $S = \{H, T\}$



Roll a die

If we are interested in the number that shows on the top face, the sample space is $S1 = \{1, 2, 3, 4, 5, 6\}$

If we are interested only in whether the number is even or odd, the sample space is $S2 = \{\text{even}, \text{odd}\}$



Task: 1

If you have a spinning wheel as shown in figure, state the sample space.

$$S=\{1,2,3,4,5,6,7,8,9,10\}$$



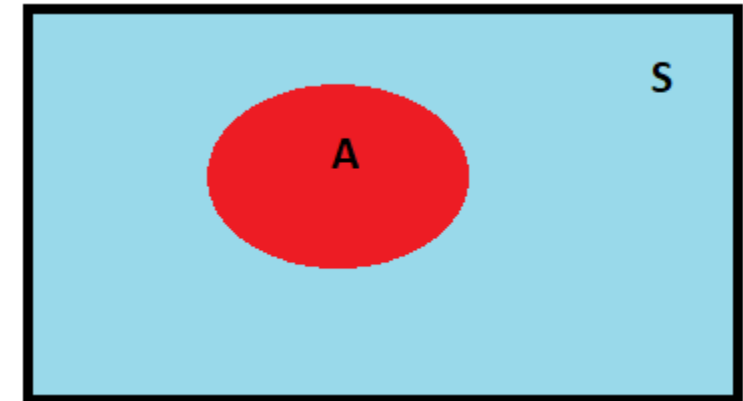
Terms

An **event** is a subset of a sample space.

The **complement** of an event A with respect to S is the subset of all elements of S that are not in A . We denote the complement of A by the symbol A' .

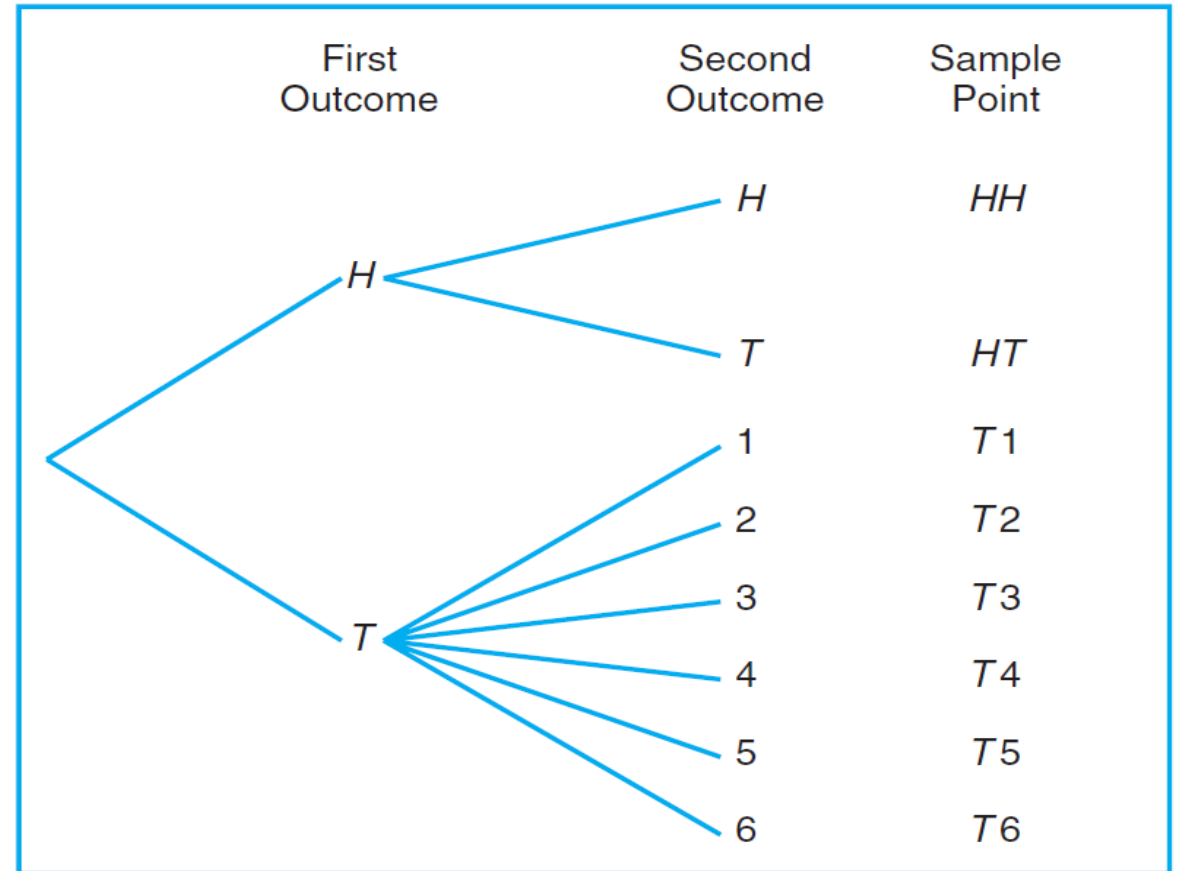
Impossible Event: no sample point, empty set.

Certain event: all sample points in sample space.



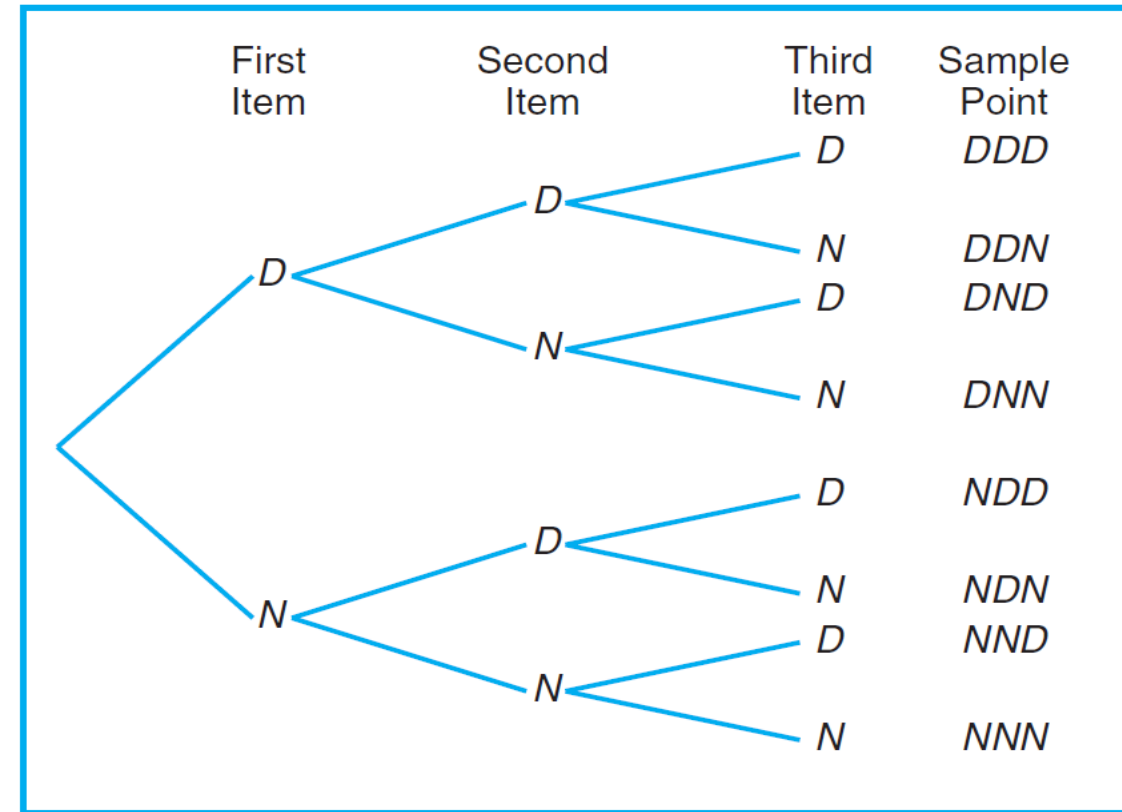
Example: 1

An experiment consists of flipping a coin and then **flipping it a second time** if a **head** occurs. If a **tail** occurs on the first flip, then a **die is tossed** once. What will be the sample space?



Task: 2

Suppose that **three items** are selected at random from a manufacturing process. Each item is inspected and classified **defective (D)**, or **non-defective (N)**. List the elements of the sample space.



Types of Sample Space

Discrete sample space (countably infinite no of sample points):

1. The no of flaws in a given length of welding
2. The no of cars crossing a toll bridge over a year

Continuous sample space (no of sample points always infinite):

1. The location on a toll bridge where a traffic accident may occur
2. Bearing capacity of a clay soil deposit between 1.5-4.0 tsf (tons per square foot)

Task

An engineering firm is hired to determine if certain waterways in Virginia are safe for fishing. Samples are taken from three rivers. (successively)

- a. List the elements of S , using F (Safe for fish) and N (Not safe for fish).

Task

An engineering firm is hired to determine if certain waterways in Virginia are safe for fishing. Samples are taken from three rivers.

- a. List the elements of E (at least two of the rivers are safe for fish).
- b. Find the probability for E.

Sol: At least two= two or more than two

- a. $E = \{FFF, FFN, FNF, NFF\}$
- b. $P(E) = 4/8$

$$P(A) = \frac{n(A)}{n(S)}$$

TASK

An engineering firm is hired to determine if certain waterways in Virginia are safe for fishing. Samples are taken from three rivers.

- a. List the elements of E (at least two of the rivers are safe for fish).
- b. Find the probability for E.

$$P(A) = \frac{n(A)}{n(S)}$$

گرتے ہیں سجدوں میں ہم
اپنی ہی حسرتوں کی خاطر اقبال
اگر گرتے صرف عشق خدا میں
تو کوئی حسرت ادھوری نہ رہتی



علامہ اقبال